E-Commerce Data Cleaning, Analysis & Reporting

Week 1 Internship Project Report

Organization: WeIntern Pvt Ltd

Prepared by: Vedant Dilip Gadade

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1. Title and Objective

Project title: E-Commerce Data Cleaning, Analysis & Reporting (Week 1).

Objective: take a raw, messy e-commerce transaction file and turn it into a clean, analysis-ready dataset, then compute the core business KPIs, examine sales trends, understand customer behaviour and present the results with visualizations and actionable recommendations.

2. Dataset Overview

The dataset contains Indian e-commerce transactions for 2024. The raw file has 5,040 rows and 11 columns.

Column	Description / observations
Order ID	Order identifier; 5,000 unique values across 5,040 rows (suggests duplicate rows)
Customer ID	1,180 unique customers
Product Name	39 unique products
Category	18 raw values due to casing variants (e.g. "APPAREL"); 6 true categories
Quantity	Units per order; 75 nulls; range -1 to 5
Unit Price	Price per unit in INR; 75 nulls; range 0 to 4,499
Total Amount	Order value; range 0 to 22,495
Order Date	Mixed formats (ISO and dd/mm/yyyy)
Payment Mode	5 modes; 101 nulls
Delivery Status	10 raw values (case variants); 100 nulls
City	15 cities

3. Cleaning Summary

The following steps were applied in order:

- Quantity nulls filled with the column median.
- Unit Price nulls filled with the median price of the same product.
- Payment Mode and Delivery Status nulls labelled "Unknown".
- Duplicate rows dropped.
- Order Date parsed to datetime using mixed-format parsing.
- Invalid records removed: Quantity <= 0 or Unit Price <= 0.
- Total Amount recomputed as Quantity x Unit Price.
- Text columns stripped and title-cased to merge casing variants.

Result: 4,955 clean rows (from 5,040), saved as cleaned_ecommerce_data.csv.

4. KPI Summary

KPI	Value
Realized Revenue (excl. Cancelled)	₹11,063,352.00
Total Orders	4,955
Average Order Value	₹2,232.77
Total Units Sold	8,149
Top Category	Electronics
Top Product	Zebronics Soundbar With Subwoofer
Unique Customers	1,180
Repeat Customer Rate	93.90%
Repeat Revenue Share	98.43%

5. Trend Analysis

Revenue was stable from January to June 2024, at roughly ₹1.86M to ₹2.09M per month (about 765 to 875 orders). January was the peak and June the lowest of the first half, a gentle decline of about 11%. From July onward revenue and order counts fall to near zero (3 to 9 orders a month).

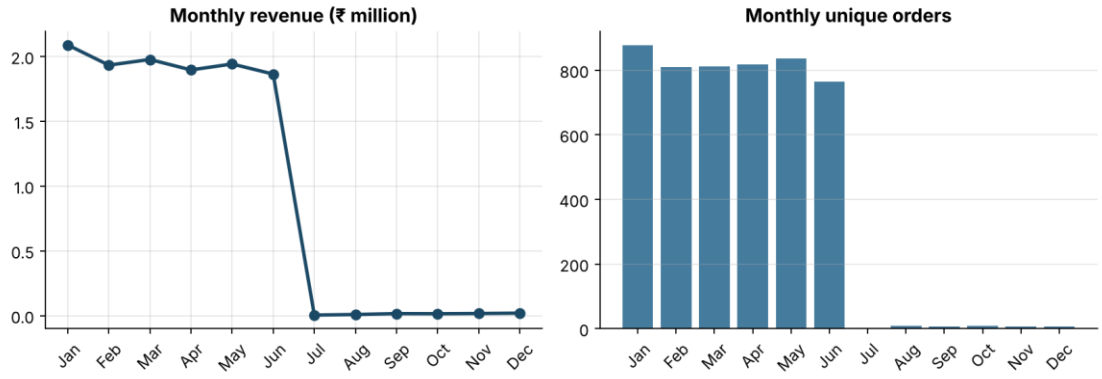


Figure 1: Monthly revenue and unique orders, 2024

Month	Revenue (₹)	Unique orders
Jan	2,088,886	876
Feb	1,932,902	809
Mar	1,976,601	812
Apr	1,895,913	817
May	1,941,758	837
Jun	1,863,761	764
Jul	4,096	3
Aug	8,418	9
Sep	15,436	6
Oct	14,405	8
Nov	16,251	7
Dec	18,988	7

Note: the July to December collapse is very likely a data artifact (see Section 10) rather than a real drop in demand.

6. Customer Insights

- 1,180 unique customers placed 4,955 orders.
- 1,108 customers (93.90%) are repeat buyers and they generate 98.43% of revenue, so the business depends heavily on returning customers.
- UPI is the dominant payment channel at 47.1% of orders, followed by Credit Card (20.0%), Cash On Delivery (11.4%), Debit Card (11.3%) and Net Banking (8.3%). Only 2.0% of orders have an unknown payment mode.
- Electronics leads category revenue (about ₹3.1M), followed by Footwear (about ₹2.5M), Apparel and Home & Kitchen (about ₹2.2M each), Fitness & Nutrition (about ₹1.2M) and Beauty & Personal Care (about ₹0.7M). Category values are read from the chart.

7. Visualization Section

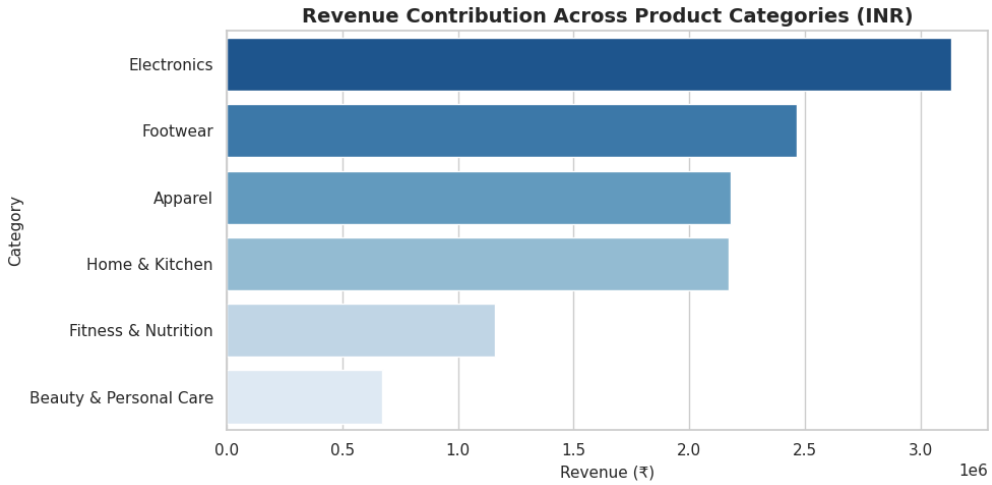


Figure 2: Revenue contribution across product categories (INR)

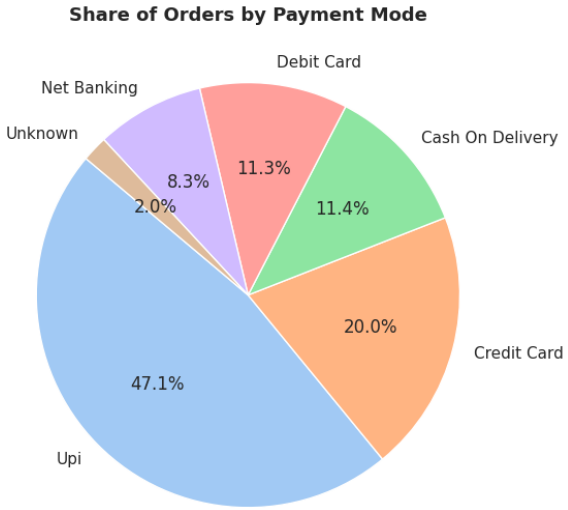


Figure 3: Share of orders by payment mode

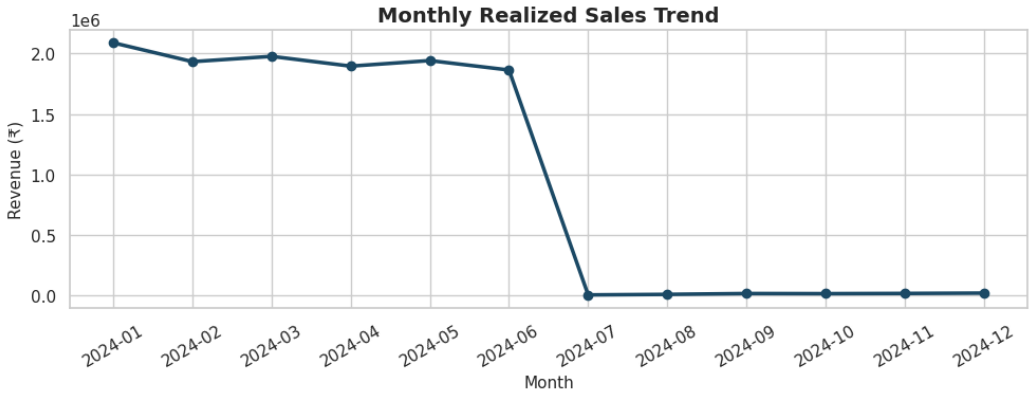


Figure 4: Monthly realized sales trend

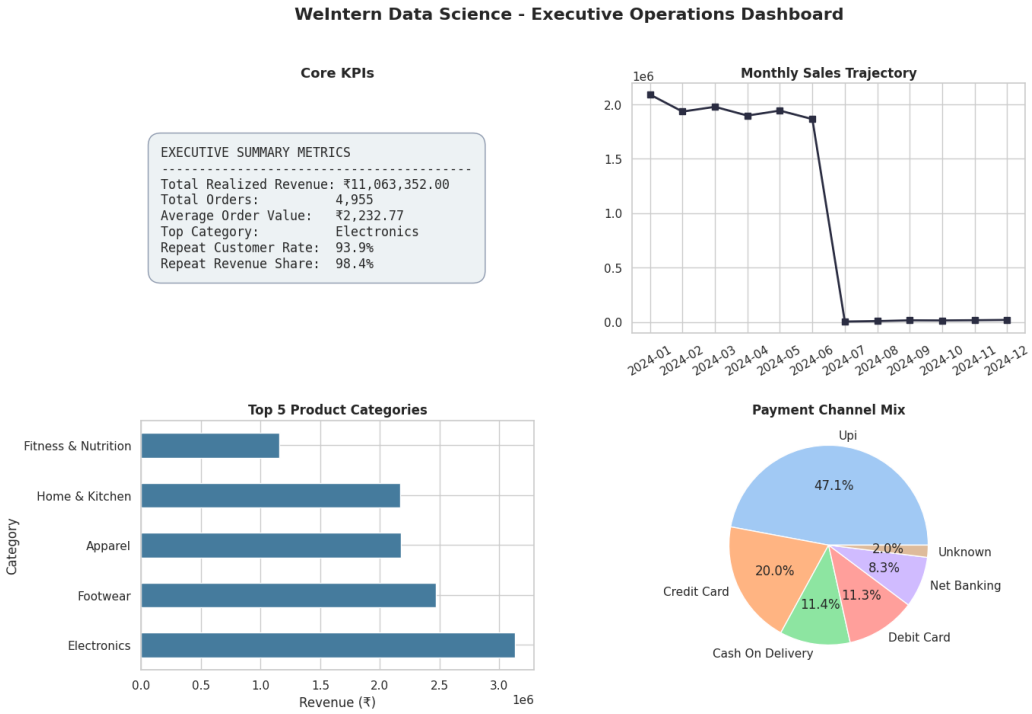


Figure 5: Executive operations dashboard

8. Key Findings

- Cleaning removed 85 rows (5,040 to 4,955) and standardized category, status and date fields, making the data analysis-ready.
- Realized revenue is ₹11.06M from 4,955 orders, with an average order value of ₹2,232.77.
- Electronics is the top category and the Zebronics Soundbar With Subwoofer is the top product.
- Sales are steady from January to June, then drop sharply from July.
- Repeat customers (93.9%) account for 98.4% of revenue.
- Nearly half of all orders (47.1%) are paid by UPI; digital methods make up the large majority.

9. Recommendations

- Launch a loyalty / rewards program to protect the repeat customers who drive almost all revenue.
- Offer one-click UPI checkout and UPI-specific offers, since UPI is the preferred method.
- Bundle electronics with accessories to raise average order value in the top category.
- Review cancellation and return patterns by city and product to find hotspots.
- Grow underperforming categories (Beauty & Personal Care, Fitness & Nutrition) with targeted promotions.
- Validate the July to December data with the source team before drawing business conclusions.

10. Limitations and Assumptions

- Missing Quantity was imputed with the median and Unit Price with the per-product median; this may distort individual orders.
- Revenue excludes Cancelled orders, while the order count and AOV denominator include them, so AOV is slightly understated.
- July to December 2024 shows only 3 to 9 orders per month. This is probably caused by date parsing (dd/mm versus mm/dd ambiguity under mixed-format parsing) or by incomplete data coverage, so the trend after June should be treated with caution.
- Order ID has 5,000 unique values against 5,040 raw rows, yet no duplicates were reported; duplicate removal ran before text normalization and may have missed some.
- The 93.9% repeat rate counts any customer with more than one order across the whole year; it is far higher than typical benchmarks and may reflect how Customer IDs were generated.
- No cost of goods sold, marketing spend or margin data is available, so profitability cannot be assessed.
- Category revenue figures in this report are read from charts and are approximate.